

The Capacitance (F) of a Capacitor in an RC Circuit

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PHY1004W

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The RC Circuit

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Introduction and Aim

An RC consists of a resistor, capacitor and, in this case, a frequency generator that drives a current through the circuit. In this arrangement, the resistor controls the charging and discharging of the capacitor. The frequency generator provides a potential difference V_{in} , with part of this applied across a resistor, denoted by V_R . The impedance (the total resistance for an RC circuit), is given by: $Z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$ (Eq (1)) where C is the capacitance of a capacitor, and $\omega = 2\pi f$, the period of oscillation provided by the frequency generator. The peak current is also provided by $I_0 = \frac{V_R}{R} = \frac{V_{in}}{Z}$ (Eq (2)). Using (1) and (2), the following equation forms: $\left(\frac{V_{in}}{V_R}\right)^2 = \left(\frac{1}{2\pi RC}\right)^2 \left(\frac{1}{f}\right)^2 + 1$ (Eq (3)) (Physics Department, 2022). The aim is thus to use multiple frequencies supplied by the frequency generator to calculate the capacitance of a capacitor and to use this mathematical model to provide some validity to the result.

Method and Apparatus

Apparatus

- A frequency generator to provide an AC current with some frequency f and potential difference V_{in}
- An oscilloscope (to measure V_{in} and V_R)
- A lunchbox consisting of a capacitor connected to a resistor, with additional connection points for the oscilloscope and function generator
- Leads to connect the oscilloscope and frequency generator to the circuit and leads to connect a circuit to the 2 channels on the oscilloscope. These leads are also used to measure the resistance
- A digital multi-meter (to calculate the resistance of the resistor)

A diagram of the apparatus used to collect data for V_{in} (V) and V_R (V)

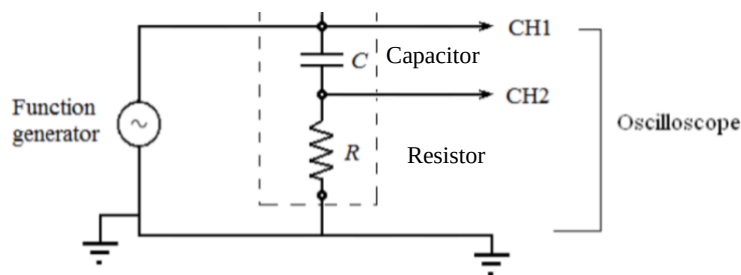


Diagram 1

(Physics Department, 2022)

A diagram of the multi-meter used to measure $R(\Omega)$

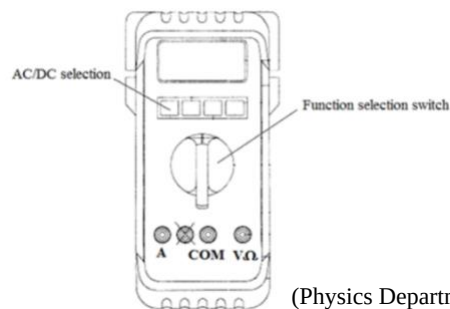


Diagram 2

(Physics Department, 2022)

Method

Firstly, 2 leads were used to connect the resistor to the multi-meter. This was done by plugging one lead in COM and the other lead into $V\Omega$. Now the other ends of the leads connect to the resistor. The function selection switch was changed to the Ω function to turn the multi-meter into an ohmmeter (Physics Department, 2022). The measurement on the multi-meter was recorded after using the range button to get a better approximation for the resistance. After this, the circuit in Diagram 1 was constructed. The oscilloscope was connected to the lunchbox such that both channels were used with channel 1 connected across the resistor and capacitor (to measure V_{in}) and Channel 2 across the resistor (to measure V_R). Next, the $V_{peak\ to\ peak}$ setting on the oscilloscope was turned on by pressing the Measure button and selecting Ch1, then using the multipurpose button to select Peak-Peak. Then the function generator is connected in series to the resistor and capacitor. The peak-to-peak voltage on the frequency generator was set to 8 V and the frequency set to any frequency value from 2 to 8 kHz. That is, $2\text{kHz} \leq f \leq 8\text{kHz}$. The oscilloscope then displays V_{in} as $V_{peak\ to\ peak}$ for channel 1 and V_R for channel 2. These values were recorded along with the period on the oscilloscope (Channels 1 and 2 read the same period). The frequency was changed, and the measurements were taken once more for a total of 8 readings.

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Data and Results

Value for R

The value for R was calculated to be: $R = (218.000 \pm 2.180) \Omega$

This was done by first measuring $R = 218.0 \Omega$.

Then this was a digital reading, meaning this is a rectangular PDF, so $u(R_{\text{resistor}}) = \frac{218.05 - 217.95}{2\sqrt{3}} = 0.02887$

Along with this, the machine is rated at 1%, so $u(R_{\text{rating}}) = 218.0 \times 1\% = 2.180$

Table 1: Showing the uncertainty budget for one resistance reading

An uncertainty budget for the resistance reading

Uncertainty Component	Standard Uncertainty (Ω)	Type of Evaluation
$u(R_{\text{rating}})$	0.02887	Type B
$u(R_{\text{resistor}})$	2.180	Type B
Combined standard uncertainty: $u_c(R) = \sqrt{0.02887^2 + 2.180^2} = 2.180$		

Yielding the result above

Tables and Graphs

A table showing the values for V_{in} (V) and V_R (V) corresponding to a measured T (s) from the oscilloscope and f (kHz) from the frequency generator

f (kHz)	$T_o(s) \times 10^{-4}$	V_{in} (V)	V_R (V)
2.372	4.220	8.08	0.630
5.616	1.778	8.08	1.45
7.951	1.252	8.08	2.04
4.619	2.164	8.08	1.28
6.560	1.526	8.08	1.76
3.121	3.204	8.08	0.900
7.439	1.346	8.08	1.96
5.069	1.970	8.08	1.40

Table 2: The raw data collected from the oscilloscope and frequency generator to use

We convert the frequency from the frequency generator to a period, by $T = \frac{1}{f}$:

A table showing the frequency (kHz) from the frequency generator as a period (s)

f (kHz)	$T(s) \times 10^{-4}$
2.372	4.216
5.616	1.780
7.951	1.258
4.619	2.165
6.560	1.524
3.121	3.204
7.439	1.344
5.069	1.973

Example calculation:

For $f = 2.372$ kHz, this is $2.372 \times 10^3 = 2372$ Hz.

Then $T = \frac{1}{f} = \frac{1}{2372} = 0.0004216 = 4.216 \times 10^{-4}$ s

Table 3: Converting frequency to a period for the frequency generator

This leads to the following ranges for T_f and T_o

The values for the period (s) from the frequency generator and the period of the oscilloscope (s), with uncertainty applied to the best estimate

$T_f(s) \times 10^{-4}$	$T_o(s) \times 10^{-4}$
$4.174 \leq T_f \leq 4.258$	$4.136 \leq T_o \leq 4.304$
$1.762 \leq T_f \leq 1.798$	$1.742 \leq T_o \leq 1.814$
$1.245 \leq T_f \leq 1.271$	$1.227 \leq T_o \leq 1.277$
$2.143 \leq T_f \leq 2.187$	$2.121 \leq T_o \leq 2.207$
$1.509 \leq T_f \leq 1.539$	$1.496 \leq T_o \leq 1.557$
$3.172 \leq T_f \leq 3.236$	$3.140 \leq T_o \leq 3.268$
$1.331 \leq T_f \leq 1.357$	$1.319 \leq T_o \leq 1.373$
$1.953 \leq T_f \leq 1.993$	$1.931 \leq T_o \leq 2.009$

Table 4: Showing the intervals for the period by the frequency generator and that displayed on the oscilloscope

Example Calculation:

There are two parts

- (i) The uncertainty of T_f : $T_f = 4.216 \times 10^{-4}$ for this example, the frequency was $f = 2.372\text{kHz}$ (as by Table 2) $f = 2372\text{Hz}$. The frequency generator was rated at 1%, so $u(f_{\text{rating}}) = 2372 \times 1\% = 23.72\text{Hz}$
This was a digital reading, so this is a rectangular PDF so $u(f_{\text{reading}}) = \frac{2372.5 - 2371.5}{2\sqrt{3}} = 0.2887\text{Hz}$

Table 5.1: *Uncertainty budget for (i)*

Uncertainty budget for this example case

Uncertainty Component	Standard Uncertainty (Hz)	Type of Evaluation
$u(f_{\text{rating}})$	23.72	Type B
$u(f_{\text{reading}})$	0.2887	Type B
Combined standard uncertainty: $u_c(f) = \sqrt{0.02887^2 + 23.72^2} = 23.72\text{Hz}$		

$$T = f^{-1} \Rightarrow u(T) = T \sqrt{\left(-\frac{u(f)}{f}\right)^2} = T \frac{u(f)}{f} = 4.216 \times 10^{-4} \frac{23.72}{2372} = 4.216 \times 10^{-6}\text{s}$$

$$\text{So } 4.174 \times 10^{-4} \leq T_f \leq 4.258 \times 10^{-4}$$

- (ii) The uncertainty of T_o : $T_o = 4.220 \times 10^{-4}$ for this example.
The oscilloscope was used and has a rating of 2%, thus $u(T_{\text{rating}}) = 4.220 \times 10^{-4} \times 2\% = 8.440 \times 10^{-6}\text{s}$
This was a digital reading, so this is a rectangular PDF, so $u(T_{\text{reading}}) = \frac{4.2205 \times 10^{-4} - 4.2195 \times 10^{-4}}{2\sqrt{3}} = 2.886 \times 10^{-8}\text{s}$

Table 5.2: *Uncertainty budget for (ii)*

Uncertainty budget for this example case

Uncertainty Component	Standard Uncertainty (s)	Type of Evaluation
$u(T_{\text{rating}})$	8.440×10^{-6}	Type B
$u(T_{\text{reading}})$	2.886×10^{-8}	Type B
Combined standard uncertainty: $u_c(T) = \sqrt{(8.440 \times 10^{-6})^2 + (2.886 \times 10^{-8})^2} = 8.440 \times 10^{-6}\text{s}$		

$$\text{Thus } 4.136 \times 10^{-4} \leq T_o \leq 4.304 \times 10^{-4}$$

From this $\left(\frac{V_{in}}{V_R}\right)^2$ and $\left(\frac{1}{f}\right)^2$ (displayed by the oscilloscope) can be calculated

A table showing the values for $\left(\frac{V_{in}}{V_R}\right)^2$ and $T_o^2(\text{s}^2)$

$T_o (\text{s}) \times 10^{-4}$	$V_{in} (\text{V})$	$V_R (\text{V})$	$\left(\frac{V_{in}}{V_R}\right)^2$	$(T_o)^2 (\text{s}^2) \times 10^{-8}$
4.220	8.08	0.630	164	17.8
1.778	8.08	1.45	31.1	3.16
1.252	8.08	2.04	15.7	1.57
2.164	8.08	1.28	39.8	4.68
1.526	8.08	1.76	21.1	2.33
3.204	8.08	0.900	80.6	10.3
1.346	8.08	1.96	17.0	1.81
1.970	8.08	1.40	33.3	3.88

Example Calculation:

For $V_{in} = 8.08$ and $V_R = 0.630$

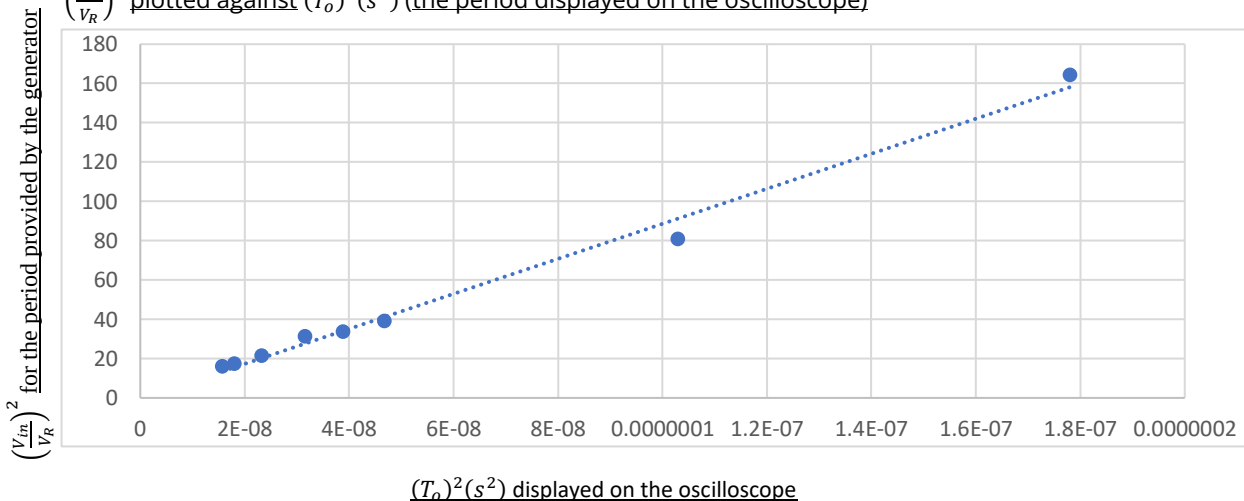
$$\left(\frac{V_{in}}{V_R}\right)^2 = \left(\frac{8.08}{0.630}\right)^2 \approx 164$$

For $T = 4.216 \times 10^{-4} \text{ kHz} = 0.4216 \text{ Hz}$

So $T^2 = 0.4216^2 = 0.178 = 17.8 \times 10^{-2} \text{ Hz}^2$

Table 6: The values of $\left(\frac{V_{in}}{V_R}\right)^2$ and $\left(\frac{1}{f}\right)^2 = T^2$ because this models a straight line with $m = \left(\frac{1}{2\pi RC}\right)^2$ and $c = 1$ by (3)

$\left(\frac{V_{in}}{V_R}\right)^2$ plotted against $(T_o)^2(\text{s}^2)$ (the period displayed on the oscilloscope)



Graph 1: $\left(\frac{V_{in}}{V_R}\right)^2$ plotted against $(T_o)^2(\text{s}^2)$ from Table 6. Using the gradient, the capacitance can be calculated and is done

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Analysis and Discussion

Calculating C using the gradient for Graph 1

By using the least squares method for Graph 1 to calculate m : $m = (89.100 \pm 3.615) \times 10^7$

Now $m = \left(\frac{1}{2\pi RC}\right)^2$ by (3) so:

$$89.100 \times 10^7 = \left(\frac{1}{(2\pi)(218.000)C}\right)^2 \Rightarrow \frac{1}{\sqrt{89.100 \times 10^7}} = 436\pi C \Rightarrow C = 2.445 \times 10^{-8} F$$

Now note that $(m)^{\frac{1}{2}} = \frac{1}{2\pi RC} \Rightarrow m^{-\frac{1}{2}} = 2\pi RC \Rightarrow \frac{1}{2\pi} m^{-\frac{1}{2}} R^{-1} = C$

Meaning,

$$u(C) = C \sqrt{\left\{-\frac{1}{2} \frac{u(m)}{m}\right\}^2 + \left\{-1 \frac{u(R)}{R}\right\}^2} = 2.445 \times 10^{-8} \sqrt{\left(-\frac{3.615 \times 10^7}{2 \times 89.100 \times 10^7}\right)^2 + \left(-\frac{2.180}{218.0}\right)^2} = 5.530 \times 10^{-10} F$$

Concluding that $C = (2.4450 \pm 0.0553) \times 10^{-8} F$

Checking the Validity of the Results

Firstly, the least squares method for Graph 1 is used to calculate c :

$$c = -0.4969 \pm 2.8029$$

Mathematically, we need $c = 1$, and we have $-3.4873 \leq c \leq 2.2121$ so this is a valid result

Now Table 3 shows all the period values shown on the oscilloscope and the period value calculated from the frequency generator. These need to agree with each other, which is the case because all the intervals for the oscilloscope overlap for some period values with the corresponding frequency generator periods. Thus, the period of the oscilloscope agrees with the period calculated from the frequency on the frequency generator

Conclusion and Recommendations

Conclusion

It was calculated that $C = (2.4450 \pm 0.0553) \times 10^{-8} F$ and that the calculated intercept is valid with the period from the oscilloscope agreeing with the period of the frequency generator, so the aim has been achieved

Discussion

Factors that may have influenced measurements include:

- Not enough digits provided by the oscilloscope. The oscilloscope provided 3 significant figures. More digits would provide a better estimate for C
- The slight difference between the period on the oscilloscope and frequency generator may slightly affect the final calculated value for C
- Lost potential difference to the resistance of the wire slightly changes the value for V_R

Sources of Uncertainty

- Reading off the display on the multi-meter, frequency generator and oscilloscope.
- The rating of the multi-meter, function generator and oscilloscope.
- The period of the oscilloscope constantly changing by small amounts
- The slight difference in the period displayed in the oscilloscope compared to the calculated value from the frequency by the frequency generator

Bibliography

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